

ABDULRAHMAN ALSAADI

MECHATRONICS & ROBOTICS ENGINEER

Jeddah, Saudi Arabia | +966 54 646 3429 | abdurahman.sa.eng@gmail.com

LinkedIn: [linkedin.com/in/abdurahmanalsaadi](https://www.linkedin.com/in/abdurahmanalsaadi) | Portfolio: [abdurahman-alsaadi.github.io](https://github.com/abdurahman-alsaadi)

PROFESSIONAL SUMMARY

Mechatronics and Robotics Engineer focused on robotics, autonomous systems, and R&D, with hands-on experience integrating robotics software, embedded control, computer vision, and mechanical design. Experienced in taking robotic systems from simulation and engineering analysis through hardware integration, testing, and physical validation.

TECHNICAL SKILLS

Robotics & Autonomy: ROS 2 Humble, Gazebo, RViz, URDF/Xacro, ArduPilot SITL, MAVLink, pymavlink, Forward & Inverse Kinematics

Embedded Systems & Control: ESP32, Raspberry Pi 5, Arduino, Cube Orange, UART, PWM, MQTT, Stepper, Servo & BLDC Motors, Systems Integration

Programming & Computer Vision: C++, Python, OpenCV, Linux (Ubuntu)

Mechanical Design, Analysis & Validation: SolidWorks, SolidWorks Simulation, FEA, Topology Optimization, DFM, Dimensional Tolerancing, Fits & Clearances, Bearing Fits, BOM Development, 3D Printing, Manufacturing Drawings

SELECTED ENGINEERING PROJECTS

WD DRONE - TEKNOFEST 2026, Rotary Wing Category

Karabük, Türkiye

Autonomous Control & Simulation

Feb 2026 - Jul 2026

- Defined system requirements and UAV architecture; contributed to component selection, BOM development, technical documentation, and technical coordination across a six-member multidisciplinary team.
- Built a custom Gazebo and ArduPilot SITL environment for autonomous waypoint search, visual target detection, target approach, payload release, and RTL; transferred the mission logic to the Raspberry Pi 5 and Cube Orange platform.
- Integrated Raspberry Pi 5, Camera Module 3, and Cube Orange through MAVLink over UART using pymavlink; implemented OpenCV color and geometric target detection at **~30 FPS**, flight-mode switching, and payload-release control.
- Validated the UAV through staged ground and flight testing, including an **8 kg** structural load test, **12-minute** unloaded hover, 7-minute hover with a **1 kg payload**, and five flight-test sessions; helped resolve severe flight oscillation through PID gain adjustment.

SCARA ROBOTIC ARM - GRADUATION PROJECT

Karabük, Türkiye

Mechatronics & Robotics Engineer

Apr 2026 - Jun 2026

- Independently designed, manufactured, assembled, and physically validated a 3-DOF SCARA robotic arm in SolidWorks with a verified **492.5 mm** reach and 100 g payload at full extension; applied DFM, dimensional tolerancing, and bearing-fit principles, and produced the BOM and manufacturing drawings.
- Applied SolidWorks Simulation, FEA, and topology optimization to redesign Arm Link 1, reducing mass by approximately **16.5%** while retaining critical load-bearing regions; designed a two-stage GT2 belt transmission, performed torque calculations, and manufactured functional PLA+ components.
- Developed the ESP32-based control system in C++ with automatic homing, forward and inverse kinematics, Cartesian point control, and programmable motion sequences; integrated the SCARA through MQTT with a vision-enabled conveyor and mobile robot for autonomous box transfer and handling.

EXPERIENCE

SMART METHODS

Jeddah, Saudi Arabia

Full Stack Robotics Engineering Intern

Jul 2025 - Aug 2025

- Completed multidisciplinary robotics training spanning mechanical design, embedded control, robot kinematics, computer vision, AI, and robotic interfaces; designed robotic-arm concepts in SolidWorks, performed FEA and torque-based motor selection, and applied Denavit-Hartenberg forward kinematics.
- Built robot-control interfaces across web and mobile platforms using PHP, MySQL, Flutter, and ESP32, enabling multi-joint control, saved motion poses, command execution, status monitoring, and HTTP/JSON-based communication; also assembled and programmed a physical 5-DOF robotic-arm training kit.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering

Jun 2021 - Jun 2026

Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- Certified SOLIDWORKS Professional - Mechanical Design (CSWP), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate - Mechanical Design (CSWA), Dassault Systèmes | May 2026
- Saudi Council of Engineers (SCE) - Engineer Membership (Active)
- ROS 2 Fundamentals, Karabük University | May 2025

LANGUAGES

Arabic - Native | English - Upper-intermediate | Turkish - Intermediate